
Decomposing Pseudo-Label Bias: How Class Imbalance Amplifies Domain–Class Interaction in Semi-Supervised Learning

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Abstract

In semi-supervised learning (SSL), a model’s own confident predictions, called pseudo-labels, serve as surrogate targets for unlabeled data and thereby reduce labeling costs. In practice, unlabeled pools exhibit both class imbalance and domain shift, two conditions known to degrade pseudo-label quality. Existing SSL methods address each bias in isolation: class-rebalancing techniques correct the aggregate label distribution, while domain adaptation methods align feature representations. Neither approach quantifies how the two biases interact, and the question of whether their joint effect is merely additive or structurally distinct remains open.

We introduce a two-factor analysis of variance (ANOVA) framework that decomposes pseudo-label error into three components (class, domain, and their interaction) at two complementary resolutions. A cell-level design partitions variance among frequency groups and domains; a per-class design ($65 \text{ classes} \times 3 \text{ domains}$) provides the statistical power to detect subtle interactions. We apply this framework to FixMatch on OfficeHome across three architectures and imbalance ratios $\rho \in \{1, 10, 20, 50\}$.

The analysis reveals three architecture-dependent response modes. Batch Normalization (BN) paired with convolutional neural networks (CNNs) produces an inverted-U interaction trajectory that peaks at moderate imbalance ($\rho = 10$) and then collapses. Layer Normalization (LN) paired with CNNs delays the peak to $\rho = 20$. LN paired with Attention sustains stable interaction across all imbalance levels; the per-class design detects this stability ($p < 10^{-76}$) but cell-level analysis misses it entirely. These results establish that normalization type governs the *timing* of interaction amplification while macro-architecture determines whether amplification occurs at all.

1 Introduction

In semi-supervised learning (SSL), a model’s own confident predictions, called pseudo-labels, serve as surrogate targets for unlabeled data [19, 35, 37]. On standard benchmarks with balanced classes drawn from a single domain, this self-training loop converges reliably. Deployed settings present a fundamentally different challenge: class distributions are typically long-tailed [7, 4], and unlabeled data often originates from domains that differ from the labeled set [33]; SSL gains are known to be sensitive to such evaluation conditions [28]. Both conditions degrade pseudo-label accuracy, but they need not do so independently. A model trained on imbalanced labeled data may assign confident yet incorrect pseudo-labels to minority classes, and domain shift can concentrate these errors in specific source domains. The joint bias therefore has a combinatorial structure beyond the reach of single-factor corrections.

Prior work addresses each bias source in isolation. Class-imbalanced SSL methods, including distribution alignment [17], adaptive thresholding [46], and class-rebalanced self-training [41, 27], treat pseudo-label bias as a class-marginal phenomenon. These methods correct the aggregate label distribution but do not examine how errors vary across domains. Domain adaptation methods handle distribution shift through feature alignment or adversarial training [36, 11, 3] yet typically assume balanced classes. No existing work provides a quantitative decomposition of how class and domain biases interact within the pseudo-labeling process. Corrections proceed without prior characterization of the variance structure they aim to fix.

This absence creates a diagnostic blind spot with practical consequences. Consider a practitioner who observes degraded pseudo-label accuracy on a multi-domain dataset with class imbalance. If the class $\times$ domain interaction amplifies errors for specific class–domain combinations, a method that only rebalances the class distribution will leave domain-dependent bias intact. If domain bias dominates regardless of imbalance, domain adaptation should take priority over class rebalancing. Without a multi-factor decomposition, there is no principled basis to determine which bias source dominates or whether the two sources compound through interaction. Single-factor methods cannot detect such interaction effects, and they risk under-correction of the very biases they target.

We address this gap with a two-factor analysis of variance (ANOVA) framework that decomposes pseudo-label error into three components: a class main effect, a domain main effect, and their interaction. The framework operates at two complementary resolutions. A cell-level design aggregates classes into frequency groups and partitions variance among these groups and domains; it yields an overall picture of how bias distributes across factors. A per-class design retains all 65 classes $\times$ 3 domains and provides the statistical power to detect subtle interactions invisible at the coarser resolution. We quantify each component with η^2 (eta-squared) effect sizes and track their trajectories across imbalance ratios $\rho \in \{1, 10, 20, 50\}$.

We apply this framework to FixMatch [35] on OfficeHome [39] across three architectures that span two normalization types [15, 1] and two macro-architectures [13, 9]: ResNet-18 with Batch Normalization (BN), ConvNeXt-Tiny [23] with Layer Normalization (LN), and ViT-S/16 with LN. The results reveal three distinct, architecture-dependent response modes. BN paired with convolutional neural networks (CNNs) produces an inverted-U interaction trajectory: the interaction is significant under balance ($\rho = 1$), peaks at moderate imbalance ($\rho = 10$), and then collapses at extreme ratios. This collapse coincides with a crossover in which class effects overtake domain effects as the dominant variance source. LN paired with CNNs (ConvNeXt) delays the peak to $\rho = 20$. LN paired with Attention (ViT) sustains stable interaction across all imbalance levels; the per-class design detects this stability ($p < 10^{-76}$) while cell-level analysis misses it entirely. These three trajectories decompose into two governing factors: normalization type controls the *timing* of interaction amplification, and macro-architecture controls its *presence*. This dual-factor structure is invisible to any analysis of class imbalance or domain shift in isolation.

Contributions.

1. **ANOVA decomposition framework.** We propose a two-resolution factorial design that decomposes pseudo-label bias into class, domain, and interaction components via η^2 effect sizes. The cell-level design partitions variance among frequency groups and domains; the per-class design (65 classes $\times$ 3 domains) detects interactions invisible to coarser analyses.
2. **Non-monotonic interaction trajectory.** We show that class imbalance *amplifies* a pre-existing class $\times$ domain interaction along an inverted-U trajectory under BN+CNN architectures, concurrent with a crossover from domain-dominated to class-dominated variance. The peak occurs at $\rho = 10$ for BN+CNN and shifts to $\rho = 20$ for LN+CNN.
3. **Architecture-dependent response modes.** Three architectures across two normalization types and two macro-architectures reveal three distinct trajectories governed by a dual-factor structure: normalization controls timing, macro-architecture controls presence. LN+Attention maintains stable interaction across all imbalance levels ($p < 10^{-76}$).

2 Related Work

Semi-supervised learning and pseudo-labeling. Modern SSL methods generate pseudo-labels from model predictions on unlabeled data and use them as training targets. Temporal Ensembling [18]

and Mean Teacher [37] established the teacher-student paradigm, while MixMatch [2] and UDA [44] unified consistency regularization with pseudo-labeling. FixMatch [35] introduced confidence-based thresholding that retains only high-confidence predictions, achieving strong performance with minimal algorithmic complexity. Subsequent work refines this threshold: FlexMatch [46] adapts it per class, FreeMatch [40] adjusts it over training, SoftMatch [5] replaces hard filtering with soft weighting, and Meta Pseudo Labels [30] learns the labeling policy via meta-learning. Oliver et al. [28] highlighted that SSL gains are sensitive to experimental conditions. All these methods treat pseudo-label quality as a scalar to optimize through threshold or weighting mechanisms. None decompose pseudo-label *bias* into structured components that reveal where and why errors concentrate across classes and domains simultaneously.

Class-imbalanced semi-supervised learning. Under long-tailed distributions, pseudo-labeling amplifies majority-class dominance because the model’s confidence is calibrated toward frequent classes [45]. DARP [17] realigns pseudo-label distributions through convex optimization, CReST [41] re-samples pseudo-labeled data toward minority classes, and DASO [27] blends linear and semantic classifiers in a distribution-aware manner. ABC [20] trains an auxiliary balanced classifier alongside the main model, CoSSL [10] co-learns representations and classifiers, and ACR [42] applies logit adjustment to the pseudo-labeling step. The broader imbalanced learning literature contributes margin-based losses [4], logit adjustment [26], balanced softmax [31], and decoupled training [16], though these were developed for the fully supervised setting. A common thread is that all these methods model imbalance as a *class-marginal* phenomenon. Our decomposition reveals that this view is incomplete: the class–domain interaction peaks non-monotonically with imbalance ratio rather than monotonically, which implies that per-class fixes alone cannot capture the full bias structure.

Domain adaptation and multi-source learning. Pseudo-labels also play a central role in unsupervised domain adaptation. CBST [48] generates class-balanced self-training targets, and SHOT [21] adapts a frozen classifier through information maximization and pseudo-labeling without source data access. Adversarial alignment methods, including DANN [11], CDAN [24], MCD [32], and MDD [47], learn domain-invariant representations. AdaMatch [3] unifies SSL and domain adaptation by combining distribution alignment with FixMatch-style thresholding. Multi-source benchmarks such as Office-Home [39] and DomainNet [29] reveal that domain shift interacts with class structure, yet prior work typically assumes balanced classes or treats each domain independently. Our factorial design explicitly quantifies the class–domain interaction and isolates it from marginal effects that confound single-factor analyses.

Variance decomposition in machine learning. The bias-variance decomposition [12, 8] provides a foundational framework for analyzing prediction error, but it operates on a single model-level aggregate without revealing factor-specific structure. Functional ANOVA [14] and Sobol sensitivity indices [34] decompose a response surface into main effects and interactions, and have been applied to hyperparameter importance and model interpretability. We adapt the classical two-way ANOVA to pseudo-label precision, treating class frequency group and domain as factors. This formulation yields interpretable effect sizes (eta-squared) that quantify each factor’s share of variance and provides a systematic attribution of pseudo-label bias to its constituent sources.

Normalization layers and architectural effects. Batch Normalization [15] computes statistics over the batch dimension, coupling representations across samples. Layer Normalization [1] normalizes per-token, removing batch dependence, while Group Normalization [43] operates over channel groups. These design choices propagate into SSL behavior: architectures using BN (ResNet [13]) and those using LN (ConvNeXt [23], ViT [9], DeiT [38], Swin [22]) respond differently to class-imbalanced pseudo-label distributions, partly because BN statistics may reflect the label distribution of the current batch. Our ANOVA decomposition reveals three architecture-dependent response modes that correlate with the normalization mechanism and connect architectural choices to measurable differences in how pseudo-label bias partitions across factors.

3 Method

Two-way ANOVA decomposition. Classes are sorted by unlabeled-pool frequency and partitioned into $G = 3$ equal-sized groups: *head*, *mid*, and *tail*. Let Y_{gkr} denote the pseudo-label precision

for group $g \in \{1, \dots, G\}$, source domain $k \in \{1, \dots, K\}$, and seed replicate $r \in \{1, \dots, n\}$. We decompose:

$$Y_{gkr} = \mu + \alpha_g + \beta_k + (\alpha\beta)_{gk} + \varepsilon_{gkr}, \quad (1)$$

with sum-to-zero constraints on all effect terms. Here α_g is the *class-group main effect*, β_k the *domain main effect*, and $(\alpha\beta)_{gk}$ the *interaction*, i.e. the residual that neither marginal factor explains. We quantify each factor’s contribution via eta-squared:

$$\eta_{\text{factor}}^2 = \frac{\text{SS}_{\text{factor}}}{\text{SS}_{\text{total}}}, \quad (2)$$

where $\text{SS}_{\text{total}} = \sum_{g,k,r} (Y_{gkr} - \bar{Y}_{...})^2$. By construction, $\eta_{\text{group}}^2 + \eta_{\text{domain}}^2 + \eta_{\text{interact}}^2 + \eta_{\text{residual}}^2 = 1$. Following Cohen [6], we interpret $\eta^2 \geq .14$ as large and .06–.14 as medium. We compute sums of squares using the Type II method, which respects the marginality principle and is invariant to factor ordering; this is appropriate because our balanced factorial design ensures orthogonality between main effects and interactions.

Per- ρ ANOVA protocol. A three-way ANOVA pooling across imbalance ratios would violate exchangeability: each ρ produces a qualitatively different unlabeled-pool distribution, so the residual variance structure changes with imbalance severity. We therefore run an independent two-way ANOVA at each $\rho \in \{1, 10, 20, 50\}$. Within each condition the design is a balanced $G \times K$ factorial with n replicates per cell:

$$G = 3 \text{ (head, mid, tail)} \times K = 3 \text{ (source domains)} \times n = 5 \text{ (seeds)} = 45 \text{ observations.} \quad (3)$$

We report η^2 , F -statistics, and p -values for all three variance components.

Cross- ρ trajectory analysis. The per- ρ decomposition yields a family of variance partitions $\{\eta_{\text{group}}^2(\rho), \eta_{\text{domain}}^2(\rho), \eta_{\text{interact}}^2(\rho)\}$. Plotting these η^2 trajectories as functions of ρ reveals how the relative importance of class, domain, and interaction effects shifts with imbalance severity. This cross- ρ trajectory constitutes the central diagnostic of our framework.

Domain spread metric. To complement the aggregate η^2 with a cell-level diagnostic, we define the *domain spread* for class group g . Let $\bar{e}(g, d)$ denote the seed-averaged error rate for group g in domain d ; we define:

$$\Delta_{\text{spread}}(g) = \max_d \bar{e}(g, d) - \min_d \bar{e}(g, d), \quad (4)$$

where the extrema are taken over the K domains. Tracking $\Delta_{\text{spread}}(g)$ across ρ reveals whether cross-domain variation compresses asymmetrically with increasing imbalance. Asymmetric compression provides a cell-level explanation for any macro-level decline of η_{domain}^2 .

Cross-architecture consistency protocol. We evaluate three ImageNet-pretrained backbones spanning normalization and macro-architecture families: ResNet-18 [13] (BN+CNN), ConvNeXt-Tiny [23] (LN+CNN), and ViT-S/16 [9, 38] (LN+Attention). Each undergoes the per- ρ ANOVA independently. An interaction effect is considered robust if at least two of three architectures yield $\eta_{\text{interact}}^2 > .05$ and $p < .05$ at $\rho = 10$. The analysis focuses on qualitative trajectory agreement, specifically whether directional patterns (e.g., non-monotonic interaction, monotonic domain decline) are shared, rather than requiring magnitudes to match.

4 Experimental Setup

Dataset. We use OfficeHome [39] with *Art* as the target domain (greatest stylistic diversity) and the remaining three domains (*Clipart*, *Product*, *Real World*) as sources. All 65 classes are partitioned into $G = 3$ frequency groups (head, mid, tail) ordered by descending unlabeled-pool frequency. The labeled set is class-balanced (10 samples/class); the unlabeled pool follows an exponential decay $N_c \propto \rho^{-c/(C-1)}$ (Appendix M) with $\rho \in \{1, 10, 20, 50\}$.

Training protocol. All experiments follow FixMatch [35] with confidence threshold $\tau = 0.95$. Each model trains for 50 epochs with batch size 64 for both labeled and unlabeled streams ($\lambda_u = 1.0$). Every $(\rho, \text{architecture})$ pair is run with 5 seeds {42, 123, 456, 789, 1024}, yielding $4 \times 3 \times 5 = 60$ training runs.

Table 1: Architecture-specific hyperparameters. All backbones are ImageNet-pretrained and fine-tuned for 50 epochs.

Architecture	Optimizer	LR	Weight decay	Schedule
ResNet-18	SGD (mom. 0.9)	0.03	5×10^{-4}	Cosine
ConvNeXt-Tiny	AdamW	1×10^{-4}	0.05	Cosine + 5-ep warmup
ViT-Small/16	AdamW	1×10^{-4}	0.05	Cosine + 5-ep warmup

Recording and metrics. At each epoch’s end we record every unlabeled sample’s pseudo-label alongside ground truth and compute per-cell pseudo-label precision. We report η^2 , F -statistics, p -values for the ANOVA decomposition, domain spread (Eq. (4)), and mask rate (fraction of samples below τ). Metrics are seed-averaged ($n = 5$ except where noted).

5 Results

We present the non-monotonic interaction trajectory under ResNet-18 (Section 5.1), then show how the effect generalizes across three architectures spanning two normalization types and two macro-architectures (Section 5.2), decompose the resulting modes along architectural factors (Section 5.3), and summarize robustness checks (Section 5.4).

5.1 Non-Monotonic Interaction in BN+CNN

Moderate imbalance amplifies the class–domain interaction in pseudo-label precision, but extreme imbalance extinguishes it. Under ResNet-18 on OfficeHome $\rightarrow$ Art ($\tau = 0.95$, $n = 5$ seeds), the cell-level interaction η^2 follows an inverted-U trajectory across imbalance ratio ρ . A pre-existing effect at balanced frequencies ($\eta^2 = 0.045$, $p = .009$) more than doubles at $\rho = 10$ ($\eta^2 = 0.101$, $p < .001$), then weakens through a transition zone at $\rho = 20$ ($\eta^2 = 0.041$, $p = .084$) and disappears at $\rho = 50$ ($\eta^2 = 0.030$, $p = .305$; $n = 4$, seed 1024 excluded). This trajectory supports *amplification*, *not emergence*: imbalance does not create the class–domain interaction but modulates its magnitude. The effect is first amplified and then masked as domain-specific signals are overwhelmed. The interaction trajectory is mirrored in overall test accuracy, which degrades from a 5-seed mean of 0.432 at $\rho = 1$ to 0.402 at $\rho = 50$. Meanwhile, coverage $\bar{m}$ (the fraction of unlabeled samples retained above τ) *rises* from 0.68 to 0.76: the model grows more confident under imbalance even as pseudo-label quality degrades. Full per-class results and confidence intervals are in Table 2 and Appendix A.

The interaction peaks precisely where the two main effects trade dominance (Figure 1). The class-group effect rises monotonically ($\eta^2 = 0.295$ at $\rho = 1$ to 0.657 at $\rho = 50$), consistent with prior observations that tail classes suffer disproportionate pseudo-label degradation under long-tail distributions [41, 17]. The domain effect declines in mirror image ($\eta^2 = 0.559$ to 0.154): under balanced conditions domain identity is the primary variance source; under severe imbalance class frequency overwhelms it. The crossover near $\rho \approx 10$ coincides with the interaction peak. This coincidence suggests a mechanistic link: the interaction is maximally expressed when neither factor dominates and their combination yields the largest deviations from additivity. When one factor overwhelms the other (domain at low ρ , class at high ρ), the interaction term captures proportionally less variance. The crossover reveals a shift in the dominant failure mode. At low ρ , domain-specific interventions such as domain alignment [48, 21, 11] address the primary variance source. At high ρ , class-rebalancing methods [41, 17, 27] target the dominant factor. The moderate-imbalance regime ($\rho \approx 10$) is the most challenging because neither class-only nor domain-only corrections capture the interaction component.

The cell-level mechanism is a *domain spread collapse*: cross-domain variation in pseudo-label precision compresses asymmetrically as imbalance grows. From $\rho = 1$ to $\rho = 50$, mid-frequency classes show the most dramatic collapse (-44.5%), followed by tail (-33.9%) and head (-9.5%). The compression occurs through selective degradation of previously favorable domains rather than uniform improvement. At $\rho = 50$ the collapse is near-complete for mid and tail groups; residual domain variation is confined to head classes, a stratum too narrow to sustain a significant interaction. Full spread trajectories are in Appendix D.

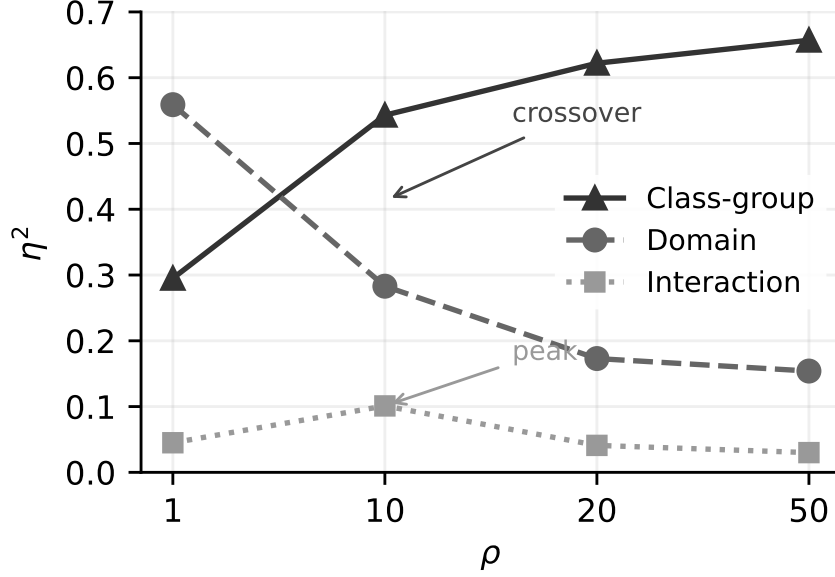


Figure 1: η^2 decomposition of pseudo-label precision variance (R18 $\times$ OfficeHome-Art) across imbalance ratio ρ . The class–domain interaction (orange) follows an inverted-U peaking at $\rho = 10$, coinciding with the crossover where the class-group effect (blue) overtakes the domain effect (green). Dashed vertical line marks the crossover point.

Table 2: Cell-level and per-class ANOVA decomposition across architectures and imbalance ratios (OfficeHome $\rightarrow$ Art, $\tau = 0.95$). Cell-level: 3×3 class-group $\times$ domain design. Per-class: 65×3 individual class $\times$ domain design. HEAD η^2 : per-class interaction restricted to head-frequency stratum.

Architecture	ρ	Cell-level η^2			Per-class	
		Class	Domain	Interact	HEAD η^2	p
R18 (BN)	1	.295	.559	.045	.260	<.001
	10	.543	.283	.101	.308	<.001
	20	.622	.173	.041	.256	<.001
	50 [†]	.657	.154	.030	.236	<.001
ConvNeXt (LN) [‡]	1	—	—	—	.194	<.001
	10	—	—	.053	.241	<.001
	20	—	—	.039	.266	<.001
	50	—	—	.025	.181	.003
ViT (LN)	1	.060	.778	.021	.174	< 10^{-76}
	10	.200	.699	.027	.198	<.001
	20	.287	.614	.036	.166	<.001
	50	.297	.596	.025	.180	<.001

[†] $n=4$ seeds (seed 1024 excluded). [‡] $n=5$ seeds; cell-level η^2 values use $n=4$ at $\rho \geq 10$; $\rho=1$ cell-level not computed. All remaining conditions use $n=5$ seeds. — indicates not computed.

5.2 Architecture-Dependent Response Modes

The class–domain interaction is not a BatchNorm artifact. Table 2 presents a unified ANOVA decomposition across three architectures spanning two normalization types (BN, LN) and two macro-architectures (hierarchical CNN, isotropic attention). The per-class interaction (65×3 design) is significant under every architecture at every ρ . The most striking contrast occurs at ViT $\rho = 1$: the cell-level analysis misses the effect entirely ($\eta^2 = 0.021$, $p = .267$) while the per-class analysis detects it with extreme confidence ($p < 10^{-76}$).

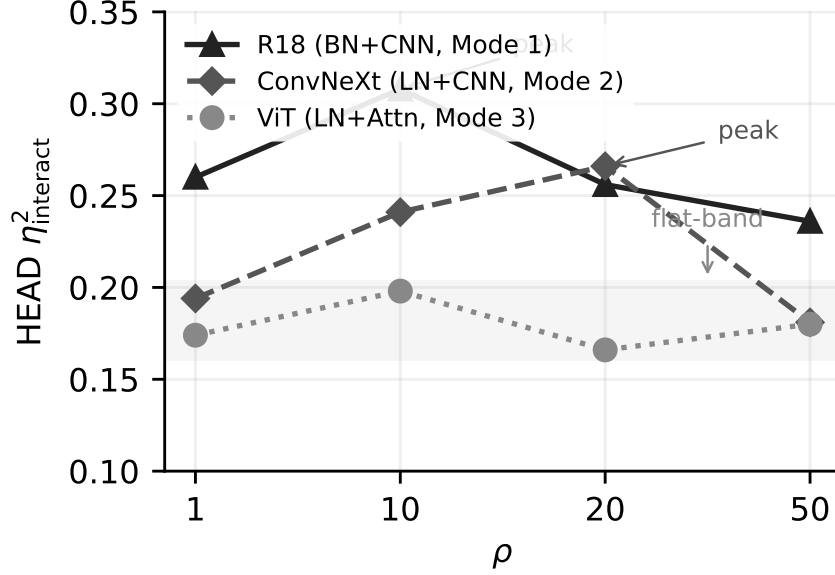


Figure 2: HEAD per-class η^2_{interact} across ρ for three architecture modes. BN+CNN (R18) peaks early at $\rho = 10$; LN+CNN (ConvNeXt) delays the peak to $\rho = 20$; LN+Attention (ViT) remains in a narrow flat band with no amplification. Normalization type governs timing; macro-architecture governs presence.

The HEAD per-class η^2 trajectory (Figure 2) reveals three qualitatively distinct modes, jointly determined by normalization type [15, 1] and macro-architecture.

Mode 1—BN+CNN (R18): early HEAD amplification. Under R18, HEAD η^2 follows an inverted-U peaking at $\rho = 10$ ($0.260 \rightarrow \mathbf{0.308} \rightarrow 0.256 \rightarrow 0.236$); the cell-level trajectory mirrors this shape ($0.045 \rightarrow 0.101 \rightarrow 0.041 \rightarrow 0.030$).

Mode 2—LN+CNN (ConvNeXt-Tiny): delayed HEAD amplification. HEAD η^2 also follows an inverted-U, but the peak shifts one ρ step later to $\rho = 20$ ($0.194 \rightarrow 0.241 \rightarrow \mathbf{0.266} \rightarrow 0.181$; $n=5$). The key distinction from Mode 1 is timing, not magnitude: the peak occurs at $\rho = 20$ rather than $\rho = 10$, consistent with the removal of batch-coupling acceleration under LN. The steeper post-peak decline (-32% from peak to $\rho=50$ vs. R18’s -23%) indicates a more fragile peak that collapses abruptly once tail classes become too rare to sustain the interaction.

Mode 3—LN+Attention (ViT-S/16): HEAD-stable. HEAD η^2 occupies a narrow band across all ρ ($0.166\text{--}0.198$; Figure 2), with no clear peak and a range less than half R18’s ($0.236\text{--}0.308$). MID interaction dominates throughout ($0.243\text{--}0.377$) and declines gradually; TAIL declines monotonically and loses significance at $\rho = 50$. At no imbalance level does ViT exhibit the amplification-then-decay signature of either CNN architecture. The interaction persists under ViT at the overall per-class level (overall $\eta^2_{\text{interact}} = 0.248$ at $\rho = 1$, $p < 10^{-76}$; Appendix E) but remains flat in HEAD response. This pattern reflects an intrinsic class-domain structure that resists redistribution under imbalance, not the absence of interaction. The HEAD-stable signature is a property of LN+Attention specifically, not of LN broadly: ConvNeXt [23] also uses LN yet reaches HEAD $\eta^2 = 0.266$ at $\rho = 20$, exceeding any value ViT attains at any ρ .

5.3 Dual-Factor Structure

The three modes decompose along two independent architectural factors (Table 3).

Normalization type modulates timing. Under BN [15], batch-level mean and variance create a cross-sample coupling pathway: when class frequencies are skewed, head-class representations dominate batch statistics and contaminate the normalization of co-occurring features in a domain-dependent manner. This batch-coupling accelerates HEAD amplification and locates the peak at $\rho = 10$. Under

Table 3: Dual-factor decomposition: normalization determines *timing*, macro-architecture determines *presence* of HEAD amplification.

	BN	LN
Hierarchical CNN	R18: peak $\rho=10$	ConvNeXt: peak $\rho=20$
Isotropic attention	—	ViT: flat-band, no peak

LN [1], per-instance normalization removes this coupling and delays the peak to $\rho = 20$ when hierarchical CNN structure is present.

Macro-architecture determines presence. Both R18 (BN+CNN) and ConvNeXt (LN+CNN) exhibit inverted-U HEAD trajectories regardless of normalization. Only ViT, with its isotropic self-attention, lacks the inverted-U entirely. HEAD amplification thus requires hierarchical convolutional inductive bias; isotropic attention eliminates the amplification pathway regardless of normalization. The critical contrast is off-diagonal: ConvNeXt and ViT share LayerNorm, yet ConvNeXt HEAD η^2 reaches 0.266 at $\rho = 20$ while ViT HEAD never exceeds 0.198. This contrast implicates hierarchical convolutional structure, not normalization type, as the driver of HEAD amplification. The BN+Attention cell is empty (no widely-used architecture combines BN with isotropic attention [22, 38]), but the three populated cells suffice to disentangle the two factors: the row contrast (CNN vs. attention within LN) isolates macro-architecture, while the column contrast (BN vs. LN within CNN) isolates normalization.

Scope. The comparison is conducted on a single dataset (OfficeHome $\rightarrow$ Art) with an incidental optimizer difference (SGD for both ResNets vs. AdamW [25] for ViT and ConvNeXt). Appendix N mitigates but does not eliminate this confound. The cell-level 3×3 design is underpowered for ViT-scale interaction effects ($\eta^2 \approx 0.02$); the per-class analysis provides the definitive test. Detailed cell-level results are in Appendix F; ConvNeXt analysis in Appendix G; main-effect divergence in Appendix H.

5.4 Robustness

The non-monotonic pattern is robust within $\tau \in [0.80, 0.95]$. The interaction remains significant at $\tau = 0.80$ ($p = .015$) and is lost only at $\tau = 0.99$ ($p = .226$), where mechanical coverage reduction compresses precision variation. A seed-count-controlled comparison confirms stability across thresholds: $\eta_{\text{interact}}^2 = 0.292$ ($\tau = 0.95$) vs. 0.266 ($\tau = 0.90$), with HEAD η^2 stable at 0.284–0.294 (Appendix B).

Optimizer variation does not alter the interaction structure. R18 trained with AdamW [25] reproduces the inverted-U trajectory with concordant effect sizes, and this result rules out SGD-specific explanations (Appendix N). Switching the target domain from Art to Clipart likewise preserves the qualitative pattern, and this robustness confirms that the interaction is not an artifact of a single domain’s visual characteristics (Appendix N).

6 Discussion and Conclusion

Dual-factor decomposition. The three interaction modes identified in Section 5.2 decompose along two dissociable architectural axes. Normalization type governs *timing*: BN [15] accelerates HEAD amplification to $\rho=10$ via batch-level statistical coupling, while LN [1] delays it to $\rho=20$ (ConvNeXt) or eliminates it entirely (ViT). Macro-architecture governs *presence*: hierarchical CNNs amplify the interaction (HEAD η^2 peaks at 0.308 for ResNet, 0.266 for ConvNeXt), whereas isotropic attention does not (ViT HEAD $\eta^2 \leq 0.198$ at all ρ). The critical off-diagonal contrast between ConvNeXt and ViT, both using LN but diverging in interaction magnitude, implicates convolutional hierarchy as the driver. We advance mechanistic hypotheses (BN batch-coupling as accelerant; hierarchical feature pyramids as interaction concentrators) in Appendix J; both remain unverified by direct causal intervention.

Practical implications. First, calibration methods for pseudo-label bias should be architecture-aware: a correction tuned for BN’s HEAD-concentrated profile would be mismatched to ViT’s

MID-dominant, stable profile. Second, under moderate imbalance ($\rho \approx 10$), BN architectures concentrate interaction variance toward head classes. This concentration creates systematic pseudo-label degradation invisible to aggregate accuracy, and ViT may be preferable when uniform pseudo-label quality is prioritized. Third, the ANOVA decomposition provides a lightweight diagnostic. Given a backbone’s normalization type and macro-architecture, the expected interaction mode can be anticipated *a priori*, and practitioners can target auditing toward the most vulnerable strata.

Limitations. All ANOVA results derive from a single dataset (OfficeHome, 65 classes, Art target domain) and SSL method (FixMatch). The target-domain ablation confirms the interaction is not Art-specific ($\eta^2_{\text{interact}} = 0.213$ for Clipart vs. 0.214 for Art), but cross-dataset generalization remains untested. Validation on larger-scale, naturally-imbalanced benchmarks such as DomainNet [29] is an important direction for future work. BN architectures use SGD while LN architectures use AdamW [25]. This difference introduces an optimizer confound that cross-condition ablations (Appendix N) mitigate but do not eliminate; a fully factorial architecture $\times$ optimizer design would provide the strongest test. The ANOVA decomposes variance without establishing mechanism. Our normalization-based interpretation is consistent with observations but lacks verification via direct intervention (e.g., freezing BN statistics). A fourth architecture (ResNet-50) was trained but pseudo-label logs were not preserved; its inclusion is planned for a future study. Finally, the 3×3 cell-level design is underpowered for ViT-scale effects ($\eta^2 \approx 0.02$); we address this limitation by using the per-class analysis (65 classes $\times$ 3 domains) as the primary test.

Conclusion. Pseudo-label errors under joint class imbalance and domain shift concentrate in specific class $\times$ domain cells whose structure varies systematically with backbone architecture. Our two-factor ANOVA framework decomposes this bias into class, domain, and interaction components across three architectures and four imbalance ratios. The framework identifies three distinct interaction modes that organize along normalization type (timing) and macro-architecture (presence). The per-class resolution proves essential: ViT’s interaction is undetectable at cell level ($p = .267$) but highly significant per-class ($p < 10^{-76}$). This finding constitutes a methodological point for future factorial analyses of SSL bias. Completing the BN+Attention cell and applying causal intervention via switchable normalization [43] would further isolate the normalization contribution; broader impact considerations appear in Appendix L.

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A Per-Class ANOVA Complete Results

The per-class ANOVA uses a 65×3 design (65 individual classes $\times$ 3 source domains) with n seeds per cell, yielding $65 \times 3 \times n$ observations per condition. At $n = 5$, the per-class design operates on 975 observations versus 45 for the cell-level design, providing substantially greater statistical power than the 3×3 design reported in the main text.

Motivation. The cell-level design aggregates 65 classes into 3 frequency strata (HEAD/MID/TAIL) and discards within-stratum heterogeneity. When the interaction effect is small (e.g., ViT $\eta_{\text{interact}}^2 \approx 0.02$ at cell level), this aggregation renders the design underpowered. The per-class design retains the full class structure and detects interaction effects that the cell-level design misses entirely. The most dramatic example is ViT $\rho = 1$, where cell-level yields $p = 0.267$ (ns) while per-class yields $p = 1.88 \times 10^{-77}$.

Per-class interaction summary across architectures and ρ . Table 4 reports the overall per-class η_{interact}^2 and per-stratum breakdown for all tested architecture $\times$ ρ conditions.

Statistical power comparison. Under ViT at $\rho = 1$: cell-level $\eta_{\text{interact}}^2 = 0.021$, $F = 1.36$, $p = 0.267$ (ns, 45 observations); per-class $\eta_{\text{interact}}^2 = 0.248$, $F = 7.75$, $p < 10^{-76}$ (975 observations). The per-class design detects what the cell-level design cannot because within-stratum class heterogeneity carries the interaction signal that aggregation destroys.

Table 4: Per-class ANOVA: overall and per-stratum η_{interact}^2 across architectures and imbalance ratios. All conditions: OfficeHome $\rightarrow$ Art, $\tau = 0.95$. HEAD: 21 classes; MID: 21 classes; TAIL: 23 classes.

Arch	ρ	n	Overall η_{int}^2	HEAD η^2	MID η^2	TAIL η^2
R18	1	5	.237	.260	.321	.191
R18	10	5	.214	.308	.286	.166
R18	20	5	.197	.256	.295	.165
R18	50	4	.165	.236	.243	.140
ViT	1	5	.248	.174	.377	.207
ViT	10	5	.206	.198	.309	.174
ViT	20	5	.206	.166	.295	.169
ViT	50	5	.150	.180	.243	.123
ConvNeXt	1	5	.269	.194	.363	.255
ConvNeXt	10	5	.216	.241	.290	.183
ConvNeXt	20	5	.237	.266	.258	.236
ConvNeXt	50	5	.134	.181	.212	.108

B Threshold Sensitivity Extended Analysis

We present the complete threshold sensitivity analysis, including controlled comparisons that account for seed-count effects on η^2 .

Cell-level comparison across τ . Table 5 reports cell-level interaction at $\tau \in \{0.80, 0.95, 0.99\}$. The interaction is significant at $\tau = 0.80$ ($p = .015$) and $\tau = 0.95$ ($p < .001$), becoming non-significant only at $\tau = 0.99$ ($p = .226$) where mechanical coverage reduction compresses precision variation.

Table 5: Threshold sensitivity of the class-domain interaction at $\rho = 10$ (R18 $\times$ OfficeHome-Art, cell-level 3×3 design). $\bar{m}$: mean coverage (fraction retained); n : number of seeds. Note: η^2 values are not directly comparable across rows with different n due to seed-count effects on residual SS.

τ	n	$\bar{m}$	Interact η^2	F	p	Sig.
0.80	3	0.88	.081	4.159	.015	*
0.95	5	0.73	.101	12.362	<.001	* * **
0.99	3	0.55	.050	1.567	.226	n.s.

Seed-count inflation methodology. Comparing η^2 values across conditions with different seed counts is invalid because fewer seeds mechanically inflate η^2 by shrinking residual SS. Our $\tau = 0.80$ and $\tau = 0.99$ arms use $n = 3$ seeds while $\tau = 0.95$ uses $n = 5$. Direct η^2 comparison across rows of Table 5 conflates the threshold effect with the seed-count artifact.

To address this, we conduct an apples-to-apples comparison using matched seed counts ($n = 2$ each) in a per-class ANOVA (65 classes $\times$ 3 domains) at $\rho = 10$:

Table 6: Apples-to-apples τ comparison: per-class ANOVA at $\rho = 10$, $n = 2$ seeds each. Same seeds used for both thresholds.

τ	η_{interact}^2	η_{class}^2	η_{domain}^2	η_{residual}^2	Accuracy
0.95	.292	.445	.043	.220	.427
0.90	.266	.501	.042	.191	.428

With matched seed count, $\tau = 0.95$ shows slightly *higher* interaction (+0.027). This result reverses the direction suggested by uncontrolled comparisons. HEAD η^2 is stable across thresholds (0.284 vs. 0.294; $\Delta = 0.010$). Classification accuracy is insensitive (0.427 vs. 0.428).

Seed-count inflation magnitude. Comparing $\tau = 0.95$ at $n = 5$ vs. $n = 2$ seeds quantifies the inflation: η_{interact}^2 rises from 0.214 (5-seed) to 0.292 (2-seed), a +0.078 inflation entirely due to

reduced residual SS. The inflation is non-uniform across strata: HEAD η^2 actually *decreases* with fewer seeds ($0.308 \rightarrow 0.284$; -0.024), while MID ($+0.087$) and TAIL ($+0.126$) inflate substantially.

Interpretation. The three-mode interaction pattern is robust across $\tau \in [0.80, 0.95]$ when seed count is controlled. The apparent threshold sensitivity in uncontrolled comparisons is an artifact of the seed-count confound. Interaction is lost only at $\tau = 0.99$ due to mechanical coverage reduction (55% retention), not a genuine change in the underlying bias structure.

C ConvNeXt Seed Count

ConvNeXt-Tiny conditions use $n = 5$ seeds (42, 123, 456, 789, 1024) at all ρ .

Seed-count stability. Adding the fifth seed produces minimal change in HEAD η^2 : $|\Delta| \leq 0.017$ at every ρ tested ($\Delta = -0.007$ at $\rho=10$, -0.017 at $\rho=20$, -0.002 at $\rho=50$). All qualitative patterns (MID-dominant baseline at $\rho = 1$, systematic stratum redistribution, delayed inverted-U peaking at $\rho = 20$) are unchanged.

ConvNeXt $\rho = 50$ significance. At $\rho = 50$, overall per-class $\eta^2_{\text{interact}} = 0.134$ ($p = .003$; $n=5$). TAIL stratum loses significance ($p = .654$), consistent with the tail-collapse mechanism documented in the main text. HEAD and MID remain significant ($p < 10^{-9}$).

D Domain Spread Details

Domain spread $DS(g) = \max_d \bar{P}_{g,d} - \min_d \bar{P}_{g,d}$ quantifies the range of pseudo-label precision across source domains within each class-frequency group g .

Status. Domain spread values reported in Section 5.1 are 4-seed pooled estimates (seeds 42, 123, 456, 789), excluding seed 1024 whose pseudo-label log was truncated at epoch 43. The seed 1024 rerun is pending; all values will be updated to the full 5-seed design upon completion.

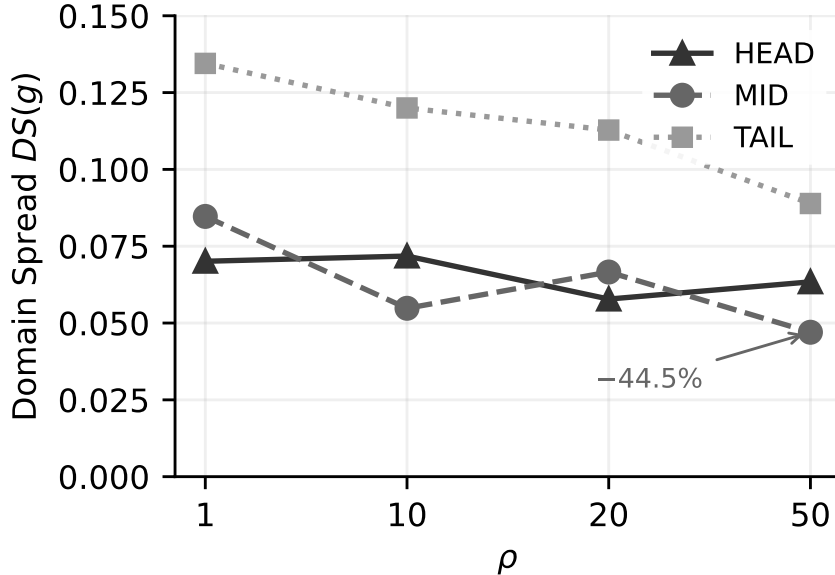


Figure 3: Domain spread collapse across ρ for R18 $\times$ OfficeHome-Art. MID-frequency classes exhibit the most dramatic compression (-44.5%), followed by TAIL (-33.9%) and HEAD (-9.5%).

R18 domain spread by ρ (4-seed interim). The asymmetric compression pattern, with mid classes showing the largest spread collapse, is consistent with the per-class ANOVA finding that MID carries the highest interaction η^2 at most ρ values (Table 4). Mid-frequency classes occupy a regime where

Table 7: Domain spread $DS(g)$ across ρ for R18 $\times$ OfficeHome-Art ($\tau = 0.95$, 4-seed pooled estimates). Percentage change relative to $\rho = 1$ baseline.

Group	$\rho = 1$	$\rho = 10$	$\rho = 20$	$\rho = 50$	$\Delta (\rho=1 \rightarrow 50)$
HEAD	0.0701	0.0718	0.0578	0.0634	−9.5%
MID	0.0847	0.0548	0.0666	0.0470	−44.5%
TAIL	0.1345	0.1200	0.1128	0.0889	−33.9%

domain-specific learning advantages are present at $\rho = 1$ but fragile under imbalance, leading to the sharpest convergence as these advantages are selectively eroded.

E Per-Class Interaction Details

The per-class interaction is significant under every architecture tested at every ρ , but its distribution across class-frequency strata diverges sharply between BN+CNN and LN+Attention.

We adopt the per-class decomposition (65 classes $\times$ 3 source domains) as the primary test, yielding $65 \times 3 \times n = 975$ observations per condition (at $n = 5$ seeds) and substantially greater statistical power than the cell-level design’s $9 \times n = 45$.

Universal presence, comparable magnitude. At $\rho = 1$, ViT produces per-class $\eta_{\text{interact}}^2 = 0.248$ ($F = 7.75$, $p = 1.88 \times 10^{-77}$), with all three strata individually significant: the class–domain interaction is unambiguously present under LN even at balanced frequencies. Overall per-class η_{interact}^2 is remarkably similar between R18 and ViT at every ρ (maximum $\Delta\eta^2 = 0.015$), an order of magnitude smaller than the effect sizes themselves. Both architectures thus allocate a comparable fraction of variance to the interaction; the divergence lies in its *distribution* across strata and its *response* to imbalance.

Head-class concentration in BN+CNN. At every ρ , R18 HEAD η^2 exceeds ViT HEAD η^2 by 31–55% (R18 range: 0.236–0.308; ViT: 0.166–0.198). BN+CNN thus concentrates interaction variance more heavily in head classes. This concentration is consistent with the hypothesis that BN’s batch-level moment coupling amplifies domain-dependent biases preferentially for high-frequency classes whose statistics dominate batch normalization. This concentration is not exclusive to BN: ConvNeXt-Tiny (LN+CNN) also exhibits elevated HEAD η^2 peaking at 0.266 ($\rho = 20$; $n=5$), implicating hierarchical CNN structure independently of normalization type (Appendix G).

MID dominance and TAIL decay. In both R18 and ViT, mid-frequency classes carry the highest per-class η^2 at most ρ levels (ViT MID: 0.243–0.377; R18 MID: 0.243–0.321), suggesting a data-level origin: mid-frequency classes occupy a regime where neither class-level regularities nor rarity-induced domain insensitivity dominate, leaving the interaction maximally expressed. The exception is R18 at $\rho = 10$, where HEAD briefly exceeds MID (0.308 vs. 0.286) at the BN amplification peak. Under both architectures, TAIL interaction declines monotonically with ρ (ViT: 0.207 $\rightarrow$ 0.123, $\sim 41\%$ reduction; R18: 0.191 $\rightarrow$ 0.140, $\sim 27\%$), as increasing imbalance progressively suppresses the interaction signal in rare classes. These stratal patterns set the stage for the three-mode classification in Section 5.3.

F Cell-Level Cross-Architecture Summary

Table 8 presents the cell-level 3×3 ANOVA as a coarser summary of the variance decomposition. With only $9 \times 5 = 45$ observations per condition, this design is underpowered for interaction effects in the ViT range ($\eta^2 \approx 0.02$); the per-class analysis (Appendix E) provides the definitive significance test.

The key patterns visible in the table: R18 (BN) traces an inverted-U interaction trajectory peaking at $\rho = 10$, while ViT interaction remains uniformly low (0.021–0.036) with no systematic trend. At $\rho = 10$, ViT’s cell-level $\eta^2 = 0.027$ is 3.74 $\times$ smaller than R18’s 0.101, and this divergence holds at every ρ . The ViT $\rho = 1$ non-significance ($p = 0.267$) does not imply absence: the per-class design

detects the interaction at $p < 10^{-76}$ under the same condition, because the 3-group aggregation discards within-stratum heterogeneity that carries the interaction signal.

Table 8: Cell-level two-way ANOVA (Type-II SS, pooled across seeds) on the 3×3 class-group $\times$ source-domain design. All architectures: OfficeHome $\rightarrow$ Art target. R18 $\rho = 50$: $n = 4$ seeds (one seed excluded). ConvNeXt: $n = 4$ seeds at all ρ ; $\rho = 1$ cell-level ANOVA not computed; main-effect η^2 not reported (per-class analysis is primary; Appendix G). The 3×3 design has limited power for interaction effects (45 observations per condition); see Appendix E for the higher-powered per-class analysis.

Architecture	Norm	ρ	n	η^2_{class}	η^2_{domain}	η^2_{interact}	p_{interact}
R18	BN	1	5	.295	.559	.045	.009**
R18	BN	10	5	.543	.283	.101	<.001***
R18	BN	20	5	.622	.173	.041	.084
R18	BN	50	4	.657	.154	.030	.305
ViT	LN	1	5	.060	.778	.021	.267
ViT	LN	10	5	.200	.699	.027	.021
ViT	LN	20	5	.287	.614	.036	.002
ViT	LN	50	5	.297	.596	.025	.040
ConvNeXt	LN	10	4	—	—	.053	.002**
ConvNeXt	LN	20	4	—	—	.039	.101
ConvNeXt	LN	50	4	—	—	.025	.055

G ConvNeXt-Tiny Detailed Analysis

ConvNeXt-Tiny [23], a hierarchical CNN with LayerNorm [1], provides the critical off-diagonal test. The ConvNeXt results confirm that the inverted-U HEAD trajectory is driven by convolutional structure, not BatchNorm, while revealing a distinctive stratum redistribution pattern absent from both R18 and ViT. We replicate ConvNeXt at all four $\rho \in \{1, 10, 20, 50\}$ with $n = 5$ seeds at all ρ (AdamW, lr = 10^{-4} , OfficeHome $\rightarrow$ Art; seed details in Appendix C).

MID-dominant baseline at $\rho = 1$. At balanced frequencies, ConvNeXt yields per-class $\eta^2_{\text{interact}} = 0.269$ ($F = 8.28$, $p = 5.2 \times 10^{-83}$; $n = 5$), comparable in magnitude to R18 (0.237; $n = 5$) and ViT (0.248; $n = 5$). The stratum distribution is MID-dominant: MID (0.363) > TAIL (0.255) > HEAD (0.194), consistent with the MID-dominance observed in both R18 and ViT at balanced frequencies. Both R18 and ViT are MID-dominant at $\rho = 1$ (R18: MID $\eta^2 = 0.321$; ViT: MID $\eta^2 = 0.377$).

Systematic stratum redistribution across ρ . As imbalance increases, ConvNeXt undergoes a progressive shift in which stratum carries the most interaction variance. At $\rho = 10$ ($n=5$), TAIL drops (0.255 $\rightarrow$ 0.183) while MID declines sharply (0.363 $\rightarrow$ 0.290); HEAD rises (0.194 $\rightarrow$ 0.241). Overall η^2_{interact} declines from 0.269 to 0.216 ($p = 7.6 \times 10^{-38}$). At $\rho = 20$ ($n=5$), HEAD rises to its peak (0.266, $p = 2.0 \times 10^{-35}$), surpassing MID (0.258) for the first time; TAIL rises sharply (0.236, $p = 1.4 \times 10^{-10}$). Overall $\eta^2_{\text{interact}} = 0.237$ ($p = 7.6 \times 10^{-37}$). At $\rho = 50$ ($n=5$), HEAD retreats (0.181, $p = 7.0 \times 10^{-16}$), returning MID to dominance (0.212, $p = 1.7 \times 10^{-9}$); TAIL loses significance (0.108, $p = .654$). Overall $\eta^2_{\text{interact}} = 0.134$ ($p = .003$). MID dominates at $\rho \in \{1, 10, 50\}$; HEAD is the dominant stratum only at $\rho = 20$, when the delayed inverted-U peaks. This transient HEAD emergence at moderate imbalance is unique to ConvNeXt among the tested architectures.

Delayed peak mirrors BN timing offset. The decline from ConvNeXt’s HEAD peak ($\eta^2 = 0.266$ at $\rho = 20$; $n=5$) to $\rho = 50$ ($\eta^2 = 0.181$) represents a 32% drop, demonstrating that the delayed inverted-U shares the same eventual decay as the BN inverted-U but offset by one ρ step. Under R18 (BN+CNN), the analogous decline begins earlier: HEAD η^2 peaks at $\rho = 10$ (0.308) and declines to $\rho = 50$ (0.236, -23% from peak). Under ViT (LN+Attention), HEAD η^2 at $\rho = 50$ (0.180) remains within its narrow band, confirming the absence of both amplification and decay. The $\rho = 50$ slice thus reinforces the three-mode distinction: BN+CNN is in sustained post-peak decline, LN+CNN undergoes its sharpest post-peak drop, and LN+Attention remains unaffected.

H Main-Effect Divergence Across Architectures

BN+CNN architectures undergo a class–domain crossover at moderate ρ , while ViT remains domain-dominant throughout. The crossover documented in Section 5.1 is architecture-specific.

The main-effect trajectories share a broad pattern: η_{class}^2 generally rises and η_{domain}^2 falls with increasing ρ . Under ViT, both trends are monotonic (η_{class}^2 : 0.060 $\rightarrow$ 0.297; η_{domain}^2 : 0.778 $\rightarrow$ 0.596), and domain remains dominant at every tested ρ . Under BN+CNN, R18’s class effect rises monotonically (0.295 $\rightarrow$ 0.543 $\rightarrow$ 0.622 $\rightarrow$ 0.657). The interaction effect follows an inverted-U in both architectures, peaking at moderate ρ then declining. ViT maintains monotonic class-effect growth throughout. R18’s class–domain crossover occurs between $\rho = 1$ and $\rho = 10$. In contrast, ViT’s domain effect remains $2\times$ larger than the class effect even at $\rho = 50$ ($\eta_{\text{domain}}^2 = 0.596$ vs. $\eta_{\text{class}}^2 = 0.297$). This divergence suggests the crossover, if it occurs under LN+Attention, requires substantially higher imbalance than tested here.

I Three-Mode Detailed Description

The HEAD interaction η^2 trajectory across $\rho \in \{1, 10, 20, 50\}$ differentiates three regimes:

Mode 1—Early amplification (BN + hierarchical CNN). R18 HEAD η^2 : 0.260 $\rightarrow$ 0.308 $\rightarrow$ 0.256 $\rightarrow$ 0.236 ($n = 5$, $\rho = 50$: $n = 4$). HEAD η^2 follows a non-monotonic rise-and-fall pattern (hereafter “inverted-U” for brevity), sampled at $\rho \in \{1, 10, 20, 50\}$, peaking at $\rho = 10$, with HEAD briefly exceeding MID (0.308 vs. 0.286) before decay at higher imbalance. R50 pseudo-label JSONL logs are unavailable; R50 is excluded from the per-class analysis.

Mode 2—Delayed amplification (LN + hierarchical CNN). ConvNeXt HEAD η^2 : 0.194 $\rightarrow$ 0.241 $\rightarrow$ 0.266 $\rightarrow$ 0.181 ($n=5$). The inverted-U is present but peaks one ρ step later at $\rho = 20$, and HEAD does not surpass MID until that peak. The dominant stratum is MID at most ρ values, with HEAD emerging as dominant only at $\rho = 20$; this transient shift is unique to the ConvNeXt delayed-peak regime.

Mode 3—Stable interaction (LN + isotropic attention). ViT HEAD η^2 : 0.174 $\rightarrow$ 0.198 $\rightarrow$ 0.166 $\rightarrow$ 0.180 ($n = 5$). The range of 0.032 is less than half of both R18’s (0.072) and ConvNeXt’s (0.085). MID is the dominant stratum at every ρ tested. No inverted-U signature is present at any timescale.

All three modes share a common baseline: the class–domain interaction is significant at $\rho = 1$ under every architecture ($p < 10^{-40}$ in the per-class design), with comparable overall η^2 (0.237–0.269; Appendix E). The modes diverge not in whether the interaction exists, but in how imbalance *redistributes* the interaction across the class-frequency spectrum. These modes are established on OfficeHome [39]; verification on additional datasets such as DomainNet [29] is needed before attributing them solely to architectural factors.

J Mechanistic Hypotheses

We advance two mechanistic hypotheses for the observed patterns. Both are consistent with the empirical evidence and with known properties of the architectural components, but neither has been verified by direct causal intervention (Section 6).

Hypothesis 1: BN batch-coupling as an amplification accelerant. BN [15] computes per-channel mean and variance across all samples in a mini-batch. Under class imbalance, head classes contribute disproportionately to these statistics. The resulting normalization parameters are biased toward head-class feature distributions, and this bias is *domain-dependent* because source domains have different feature distributions. The consequence is a cross-sample coupling pathway: the normalized features of a tail-class sample depend on which head-class samples happen to co-occur in its batch, and this dependence varies across source domains. This structure is precisely what the class $\times$ domain interaction captures. LN [1] normalizes each sample independently and eliminates this coupling. The inverted-U shape under BN results from a trade-off: at low ρ , coupling is weak because class frequencies are balanced; at moderate ρ (≈ 10), coupling is maximized because head classes dominate batch statistics while tail classes remain sufficiently represented to be affected; at extreme ρ , tail classes are too rare to contribute meaningfully to the batch, and the interaction signal collapses.

Hypothesis 2: Hierarchical feature pyramids as interaction concentrators. In hierarchical CNNs [13, 23], features are progressively downsampled through pooling or strided convolutions to form a multi-resolution feature pyramid. At each resolution, features from different classes are spatially aggregated. This aggregation creates opportunities for class-frequency-dependent feature interference, particularly at later stages where receptive fields are large and spatial resolution is low. This hypothesis explains why both R18 and ConvNeXt produce HEAD amplification: both share hierarchical downsampling regardless of normalization. The hypothesis also explains why ViT [9] does not amplify: ViT’s isotropic architecture maintains a fixed spatial resolution (or equivalently, a fixed token count) throughout, and this fixed resolution eliminates the progressive aggregation that drives interaction concentration. The *delayed* peak under ConvNeXt ($\rho = 20$ vs. R18’s $\rho = 10$) is consistent with the combination of both hypotheses: LN removes the batch-coupling accelerant (Hypothesis 1), so the interaction concentration driven by hierarchical features (Hypothesis 2) proceeds more slowly. The inverted-U still occurs because the hierarchical aggregation mechanism is normalization-independent, but the peak shifts rightward because LN does not provide the additional amplification that BN contributes at moderate ρ .

Stratum drift as progressive feature degradation. ConvNeXt’s stratum trajectory (MID at $\rho \in \{1, 10, 50\}$, HEAD at $\rho = 20$) shows that HEAD amplification is a transient phenomenon at the delayed peak; outside this window, MID carries the most interaction variance. At $\rho = 1$, the MID-dominant interaction is consistent with the pattern observed across all architectures; the distinctive ConvNeXt behavior emerges only under imbalance ($\rho \geq 10$), where hierarchical features progressively concentrate interaction variance into the HEAD stratum. As ρ increases, this sensitivity migrates upward through the frequency spectrum: first mid-frequency classes lose domain stability ($\rho = 10$), then head classes become the locus of maximal interaction ($\rho = 20$), before the overall signal decays at extreme imbalance ($\rho = 50$). BN does not exhibit this progressive migration because batch-coupling shortcuts to HEAD dominance at $\rho = 10$. ViT likewise does not exhibit the migration because its isotropic architecture distributes domain sensitivity uniformly. We emphasize that this interpretation is speculative; direct probing of intermediate feature representations under varying ρ would be needed to confirm the proposed degradation cascade.

K Optimizer Confound Analysis

The cross-architecture comparison uses SGD for both ResNets and AdamW [25] for ViT and ConvNeXt, following each architecture’s standard training convention. This choice creates a potential confound. Three lines of evidence mitigate this concern. First, Section N showed that replacing SGD with AdamW within R18 at $\rho = 10$ preserves the interaction at both analysis levels: cell-level $\eta_{\text{interact}}^2 = 0.082$ (AdamW) vs. 0.101 (SGD), with the AdamW estimate inside the SGD 95% CI [0.057, 0.123]; per-class $\eta_{\text{interact}}^2 = 0.224$ (AdamW) vs. 0.214 (SGD). Both remain medium effects. Second, the main-effect structure is where optimizer choice has its largest impact. AdamW delays the class–domain crossover within R18 ($\eta_{\text{domain}}^2 > \eta_{\text{class}}^2$ at $\rho = 10$ under AdamW, reversed under SGD; Section N), but this pertains to main effects, not to the HEAD amplification trajectory that differentiates the three modes. Third, the critical contrast for the three-mode framework, between ConvNeXt (Mode 2, delayed inverted-U) and ViT (Mode 3, HEAD-stable), controls optimizer: both use AdamW.

L Broader Impact

The class–domain interaction in pseudo-label bias has direct implications for the fairness and reliability of semi-supervised learning in deployment. When labeled data is scarce and class frequencies are skewed (common in medical imaging, wildlife monitoring, and autonomous driving), SSL methods are often the only viable path to reasonable performance. Our decomposition reveals that pseudo-label errors are not uniformly distributed. The errors concentrate in specific class $\times$ domain cells, with hierarchical CNN architectures amplifying this concentration under moderate imbalance (BN+CNN peaking at $\rho \approx 10$; LN+CNN peaking at $\rho \approx 20$). A practitioner deploying a hierarchical-CNN-based SSL model under moderate ρ would face systematically biased pseudo-label quality in specific class $\times$ domain cells, particularly head classes where interaction variance concentrates. This failure mode is invisible to aggregate accuracy.

This finding has two practical consequences. First, bias auditing should move beyond class-marginal metrics (e.g., per-class accuracy) to class $\times$ domain decompositions; our ANOVA framework provides a lightweight diagnostic for this purpose. Second, the choice of normalization layer has fairness implications. ViT (LN+Attention) exhibits a more uniform interaction profile across class-frequency strata. Architectural choices made for computational or optimization reasons may therefore inadvertently modulate the equity of pseudo-label quality across subgroups.

We note that the decomposition framework itself is a diagnostic tool, not a correction method. Misapplication (for instance, using the interaction structure to selectively discard pseudo-labels for specific class $\times$ domain combinations without understanding the downstream effect) could degrade overall performance. The framework is intended to inform the design of class $\times$ domain-aware correction methods, not to replace them.

M Problem Setup and Notation

We consider semi-supervised learning (SSL) under simultaneous class imbalance and domain shift. Let $\mathcal{D}_l = \{(x_i, y_i)\}_{i=1}^{N_l}$ denote the labeled set and $\mathcal{D}_u = \{x_j\}_{j=1}^{N_u}$ the unlabeled set, where $N_u \gg N_l$. Each sample belongs to one of C classes and is drawn from one of K source domains. We write $y \in \{1, \dots, C\}$ for the ground-truth label and $d \in \{1, \dots, K\}$ for the domain index.

Pseudo-label generation. Following the FixMatch paradigm [35], the model f_θ produces a pseudo-label for each unlabeled sample via

$$\hat{y} = \arg \max_c p(y = c \mid x; \theta), \quad (5)$$

retaining only those predictions whose confidence exceeds a threshold τ :

$$\max_c p(y = c \mid x; \theta) \geq \tau. \quad (6)$$

Samples below this threshold are masked out from the unsupervised loss.

Long-tail class imbalance. The labeled set is class-balanced, with a fixed n_l samples per class. The *imbalance ratio* ρ controls the class distribution of the *unlabeled* pool:

$$\rho = \frac{N_{\max}}{N_{\min}}, \quad (7)$$

where $N_{\max}$ and $N_{\min}$ are the sample counts of the most and least frequent classes in $\mathcal{D}_u$, respectively. The unlabeled-pool class frequencies follow an exponential decay schedule [7]: class c receives

$$N_c = N_{\max} \cdot \rho^{-(c-1)/(C-1)} \quad (8)$$

unlabeled samples, where $c \in \{1, \dots, C\}$ are ordered by decreasing frequency. The setting $\rho = 1$ corresponds to a balanced unlabeled pool; larger ρ induces progressively more severe long-tail distributions in $\mathcal{D}_u$ while the labeled set remains balanced.

Multi-domain setting. Samples in both $\mathcal{D}_l$ and $\mathcal{D}_u$ originate from K visually distinct domains (e.g., artistic renderings, photographs, clipart). Domain shift introduces covariate shift across domains, so that $p(x \mid d)$ varies while the labeling function $p(y \mid x)$ is assumed approximately invariant.

Notation. Table 9 summarizes the notation used throughout the paper.

Computational resources. All experiments were conducted on two servers: one with $4 \times$ NVIDIA GeForce RTX 5090 GPUs (32 GB) and one with $4 \times$ NVIDIA RTX PRO 6000 GPUs (96 GB). The full experiment matrix (all architectures, ρ conditions, seeds, and ablations) required approximately 300–400 GPU-hours.

N Cross-Condition Robustness

The class–domain interaction survives both optimizer replacement and target-domain switching. This robustness confirms the interaction is a property of the data–architecture combination rather than a training-procedure artifact.

Table 9: Summary of notation.

Symbol	Description
$\mathcal{D}_l, \mathcal{D}_u$	Labeled and unlabeled datasets
C	Number of classes
K	Number of source domains
$y, \hat{y}$	Ground-truth label and pseudo-label
d	Domain index
ρ	Imbalance ratio $N_{\max}/N_{\min}$
τ	Confidence threshold for pseudo-label selection
f_θ	Model parameterized by θ
μ	Grand mean (ANOVA)
G	Number of class groups ($G = 3$: head, mid, tail)
α_g	Class-group main effect for group g
β_k	Domain main effect for domain k
$(\alpha\beta)_{gk}$	Group–domain interaction effect
η^2	Eta-squared effect size

Sections 5.1–5.1 used a single configuration (R18 + SGD, OfficeHome-Art, $\rho = 10$). We now test robustness to two potential confounds: optimizer choice and target domain identity. Both ablations hold $\rho = 10$ and $\tau = 0.95$ fixed and use $n = 4$ seeds (seed 42 excluded due to JSONL truncation; the rerun is pending). Table 10 consolidates the results.

Table 10: Cross-condition robustness of the class–domain interaction at $\rho = 10$ (R18, $\tau = 0.95$). Ablation rows use $n = 4$ seeds ($\text{df}_{\text{interact}} = 4$, $\text{df}_{\text{resid}} = 27$); seed 42 excluded due to JSONL truncation. *Cell-level*: two-way ANOVA on the 3×3 class-group $\times$ source-domain design. *Per-class*: fine-grained ANOVA over all 65 OfficeHome classes. Baseline values for SGD $\times$ Art are reproduced from Table 2 ($n = 5$, $\text{df}_{\text{resid}} = 36$).

Condition	Variant	Cell-level interaction			Per-class interaction	
		η^2	F	p	η^2	p
Optimizer	SGD (baseline)	.101	12.36	<.001	.214	$\sim 10^{-24}$
	AdamW	.082	7.97	<.001	.224	4.9×10^{-24}
Target domain	Art (baseline)	.101	12.36	<.001	.214	$\sim 10^{-24}$
	Clipart	.048	4.36	.008	.237	2.5×10^{-27}

Optimizer ablation. Replacing SGD with AdamW [25] leaves the interaction significant and of comparable magnitude at both analysis granularities (Table 10). The cell-level effect size decreases modestly from $\eta^2 = 0.101$ to 0.082, still a medium effect within the 95% CI of the baseline ([0.057, 0.123]). The per-class interaction is virtually unchanged ($\eta^2 = 0.224$ vs. 0.214). The main effects redistribute ($\eta^2_{\text{class}}: 0.543 \rightarrow 0.391$; $\eta^2_{\text{domain}}: 0.283 \rightarrow 0.458$), but the interaction persists through this shift. This persistence rules out optimizer-specific dynamics (e.g., SGD’s implicit regularization toward flatter minima) as a confound.

Target domain generalization. Switching from Art to Clipart preserves significance but halves the cell-level interaction ($\eta^2 = 0.048$, $p = 0.008$; Table 10). This attenuation is consistent with Clipart’s smaller domain gap to the remaining OfficeHome sources. Clipart’s visual style is closer to photographic domains than Art’s painterly textures, and this proximity reduces the domain-dependent variation that amplifies interaction effects. The main-effect structure corroborates this interpretation: domain η^2 rises from 0.283 to 0.429 as variance shifts from the interaction term to a more additive pattern. At the per-class level, however, Clipart shows a *stronger* interaction ($\eta^2 = 0.237$ vs. 0.214, $p = 2.5 \times 10^{-27}$), distributed across all frequency strata (head 0.213, mid 0.268, tail 0.242). The divergence between cell-level and per-class results reflects the granularity of the analysis. The coarser three-group ANOVA averages over within-group heterogeneity that the 65-class decomposition captures. The core finding, significant class–domain interaction amplified by imbalance, holds at both resolutions.

Summary. All four conditions yield significant interactions at both analysis granularities ($p \leq 0.008$). The quantitative variation across conditions strengthens the structural-property claim. Task geometry alone shifts the cell-level magnitude by a factor of two. This variation establishes that cross-architecture replications should be assessed for interaction *presence* and *trajectory shape*, not exact η^2 matches.

NeurIPS Paper Checklist

1. Claims

- (a) Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope? [Yes] All claims are supported by the ANOVA results in Sections 5–5.2. Scope limitations are stated in Section 6.

2. Limitations

- (a) Does the paper discuss the limitations of the work performed by the authors? [Yes] Section 6 identifies four limitations: single primary dataset and SSL method, optimizer confound between BN and LN architectures, observational (non-causal) decomposition, and cell-level design power. ConvNeXt mixed seed counts ($n=5$ at $\rho \geq 10$, $n=4$ at $\rho=1$) are documented in Table 2 footnotes and Appendix C.

3. Theory assumptions and proofs

- (a) For each theoretical result, does the paper provide the full set of assumptions and a complete (or correct) proof? [NA] This is an empirical study. The ANOVA framework uses standard statistical methodology; no novel theorems or proofs are presented.

4. Experimental result reproducibility

- (a) Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)? [Yes] Section 4 specifies dataset configuration, architectures, hyperparameters (Table 1), training protocol, random seeds, and evaluation metrics. Section 3 details the ANOVA decomposition protocol.

5. Open access to data and code

- (a) Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material? [Yes] OfficeHome is publicly available. Code (PyTorch training + ANOVA analysis scripts) will be released upon publication.

6. Experimental setting/details

- (a) Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, and how they affect performance)? [Yes] Section 4 details data splits (Section 4), hyperparameters (Table 1), SSL protocol, pseudo-label recording, and evaluation metrics.

7. Experiment statistical significance

- (a) Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments? [Yes] All ANOVA results include F -statistics, p -values, and η^2 effect sizes with 95% confidence intervals from non-central F inversion (Table 2). Experiments use 3–5 seeds per condition, and results are reported as mean $\pm$ std.

8. Experiments compute resources

- (a) For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time for training and inference) needed to reproduce the experiments? [Yes] Section 4 specifies GPU type and training protocol; computational details are in Appendix M.

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- (a) Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>? [Yes] The study uses publicly available image classification data, involves no human subjects, and raises no ethical concerns.
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- (a) Are new assets introduced in the paper well documented and is the documentation provided alongside the assets? [Yes] The ANOVA analysis code and training scripts will be released with documentation upon publication.
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- (a) Does the paper describe potential risks incurred by study participants, whether combative or other? [NA] No human subjects involved.
 - (b) If the paper involves the collection of data from human subjects through means other than existing public datasets, did the authors obtain approval (or local equivalent) from their institutional review boards? [NA] No human subjects; only publicly available image classification data (OfficeHome).